# Cybersecurity Incident Report: Network Traffic Analysis

| Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log. |
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| The UDP protocol reveals that:<br>My computer tried to contact the DNS server at 203.0.113.2 to get the IP address for yummyrecipesforme.com, but the request did not go through. The request was sent using UDP. Since DNS did not return an IP address, the browser could not load the website.<br>This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message: udp port 53 unreachable. I saw this same error each of the three times the request was tried.<br><br>The port noted in the error message is used for: DNS<br><br>The most likely issue is:The DNS server was not responding to requests on port 53. It looks like the DNS service was not running or was not working correctly. |

| Part 2: Explain your analysis of the data and provide at least one cause of the incident. |
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| Time incident occurred:1:24 pm based on the first timestamp in the tcpdump log.<br><br>Explain how the IT team became aware of the incident:Several customers reported that they could not get to www.yummyrecipesforme.com. They received a “destination port unreachable” error after waiting for the page to load.<br><br>Explain the actions taken by the IT department to investigate the incident: First, I tried to visit the website and got the same error. Next, I used tcpdump to capture the network traffic while I tried to load the page again. I looked through the log and saw the UDP requests and ICMP error messages. The issue was then reported to my supervisor, and the security engineers are now working on it.<br><br>Note key findings of the IT department's investigation (i.e., details related to the port affected, DNS server, etc.): My computer’s IP address is 192.51.100.15, and the DNS server’s |

IP address is 203.0.113.2.

My computer used UDP to ask the DNS server for the website's IP address.

The response was an ICMP error that said "udp port 53 unreachable."

The error showed up all three times.

Port 53 is the port used for DNS, so the problem appears to be with the DNS service.

The browser could not get the website's IP address, so the page could not load.

Note a likely cause of the incident:The DNS service may have stopped running, or it may have been set up incorrectly, so nothing was listening on port 53. I think the next step would be to check whether the DNS service is running and restart or fix it if needed. I cannot tell from this log whether an attack caused the problem, so that would need more investigation.